



Jarn de Jong

jarn@jarndejong.nl Jarndejong.nl

Quantum Physicist,
Cryptographer, Developer



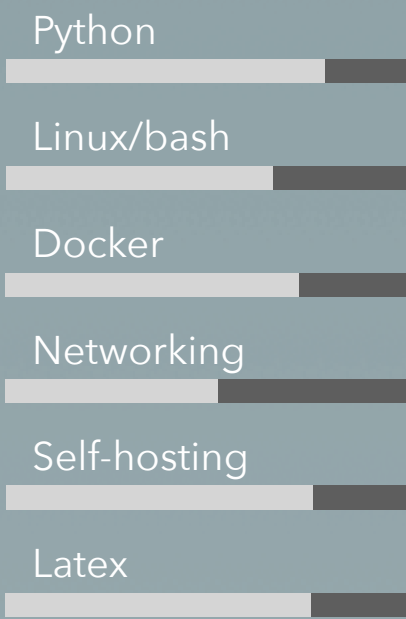
List of publications

Bio

Quantum physicist with a strong background in a wide range of topics. Software developer & tech enthusiast with a strong track record in Python, self-hosting, Docker and more.

Phd (summa cum laude) in quantum communication & cryptography, BSc. & MSc. in Applied Physics.

Programming & Computer skills



Quantum skills

- Networking & communication
- Stabilizer & graph states
- Multipartite entanglement
- Local Equivalence
- QKD & CKA
- Anonymity in quantum networks
- Photonic systems
- Distributed systems
- QECC & Fault-tolerance
- State- and Process tomography
- Quantum Internet use cases

Languages

Dutch	Native
English	C2
German	B1-B2

Experiences

2026-		PsiQuantum	System architect, quantum networking
Developing tools and methods for modelling, defining and simulating photonic quantum systems.			
2022-2025		Quantum Internet Alliance	Use Case team member
Project management, use case development and simulation, reporting.			
May 2023		LIP6 (Sorbonne Université)	Visiting researcher
Research visit at the LIP6 team. Present earlier work and collaborate with multiple researchers.			
2020-2025		Technische Universität Berlin	Scientific Employee
Designing & giving lectures, student supervision, scientific talks.			
2019-2020		QuTech	Research assistant & software developer
Hardware- and software development, simulations. Work span off into Q*Bird.			
2019		TNO	Quantum Technology Intern
Design and develop experiments for upcoming quantum device. Software development.			
2014-2015		Gymnasium Sorghvliet	Physics teacher intern
Half-year teaching internship for high school physics. Culminated in official qualification.			

Education and qualifications

2020-2024		TU Berlin, DE	PhD (Dr. Rer. Nat.)	Summa cum laude
Thesis under Anna Pappa in her quantum communication & cryptography group.				
2016-2019		TU Delft, NL	MSc. Applied Physics	GPA: 8.4
Quantum technologies track, <u>thesis</u> under Barbara Terhal at QuTech.				
2014-2015		TU Delft, NL	High school teaching qualification	N/A
High school physics teaching qualification with half-year internship at Gymnasium Sorghvliet during minor of BSc.				
2012-2015		TU Delft, NL	BSc. Applied Physics	GPA: 7.5
Thesis under prof. M. Verhaegen at Delft Centre for Systems and Control.				

Personal projects and extracurricular activities

Software projects	
Various software projects, e.g. the <u>Graphstabilizer</u> package and <u>Spotify2PlexPlaylistSyncer</u> .	
Self-hosting	
Deploying, hosting & orchestrating <u>my website</u> , multiuser <u>Nextcloud</u> -, Git-, media- and photo servers, and many more.	
2011-2025	Tutor, teaching (assistant), designing courses
High school tutoring, being teaching assistant, giving retake courses, designing and giving MOOC's.	

Links

Published	arXiv
---------------------------	-----------------------

Published	arXiv
---------------------------	-----------------------

Published	arXiv
---------------------------	-----------------------

Published	arXiv
---------------------------	-----------------------

Published	arXiv
---------------------------	-----------------------

Published	arXiv
---------------------------	-----------------------

arXiv

Published	arXiv
---------------------------	-----------------------

Published	arXiv
---------------------------	-----------------------

Slides

Slides

Slides

Slides

Slides

Slides	Gif
------------------------	---------------------

Publications

2020	<i>PRX Quantum</i> 1, 020325	<i>F. Hahn, J. de Jong, A. Pappa</i>
<i>Anonymous Conference Key Agreement</i>		
2021	<i>New J. Phys.</i> 23 083026	<i>C. Thalacker, F. Hahn, J. de Jong, A. Pappa, S. Barz</i>
<i>Anonymous and secret communications in quantum networks.</i>		
2022	<i>PRX Quantum</i> 3, 040306	<i>F. Grasselli, G. Murta, J. de Jong, F. Hahn, D. Bruß, H. Kampermann, A. Pappa</i>
<i>Secure Anonymous Conferencing in Quantum Networks</i>		
2023	<i>Quantum</i> 7, 1117	<i>J. de Jong, F. Hahn, J. Eiser, N. Walk, A. Pappa</i>
<i>Anonymous conference key agreement in linear quantum networks</i>		
2023	<i>Phys. Rev. Research</i> 5, 033222	<i>L. Rückle, J. Budde, J. de Jong, F. Hahn, A. Pappa, S. Barz</i>
<i>Experimental ACKA using linear cluster states</i>		
2024	<i>Phys. Rev. Research</i> 6, 013330	<i>J. de Jong, F. Hahn, N. Tcholtchev, M. Hauswirth, A. Pappa</i>
<i>Extracting GHZ states from linear cluster states</i>		
2024	(preprint) <i>arXiv:2410.03961</i>	<i>A. Burchardt, J. de Jong, L. Vandr�</i>
<i>Algorithm to Verify Local Equivalence of Stabilizer States</i>		
2025	<i>Phys. Rev. A</i> 111, 052449	<i>L. Vandr�, J. de Jong, F. Hahn, A. Burchardt, O. G�hne, A. Pappa</i>
<i>Distinguishing graph states by the properties of their marginals.</i>		
2025	<i>Phys. Rev. Applied</i> 24, 054053	<i>J. de Jong, S. Scheiner, N. Solomons, Z. Chaoui, D. Markham, A. Pappa</i>
<i>Anonymous and private parameter estimation in networks of quantum sensors</i>		

Selected talks

2021	<i>PhD Seminar</i>	<i>Berlin School of Optical Sciences and Quantum Technologies</i>	Online
<i>Measurement-based and blind quantum computation</i>			
2022	<i>Research visit</i>	<i>Koch group (Freie Universit�t Berlin)</i>	Berlin, DE
<i>Graph states as quantum networks</i>			
2023	<i>Conference</i>	<i>DPG spring meeting 2023, atomic physics & quantum information</i>	Hannover, DE
<i>Extracting maximal entanglement from linear cluster states</i>			
2023	<i>Research visit</i>	<i>LIP6, Sorbonne Universit�</i>	Paris, FR
<i>Extracting GHZ states from linear cluster states - to extract or not to extract</i>			
2023	<i>Meeting</i>	<i>Yearly Quantum Internet Alliance meeting</i>	Paris, FR
<i>Anonymity in quantum network protocols</i>			
2024	<i>Invited speaker</i>	<i>MATH+ Einstein workshop on Quantum Cryptography and Quantum Networks</i>	Berlin, DE
<i>Quantum Networking using graph theory</i>			